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Full state quaternion-based observer control for multirotor aerial grasping

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***Abstract*— This paper presents an enhanced observer control strategy for multirotor aerial grasping. Unlike previous approaches, which focused solely on translational dynamics, this method incorporates dual observers—one for the translational subsystem and another for the rotational dynamics. By leveraging quaternions, the proposed control framework provides a singularity-free representation of orientation while naturally decoupling rotational and translational dynamics. This allows the system to be treated as fully actuated in both position and orientation, improving disturbance rejection and compensating for torques induced by off-center or asymmetrically shaped objects during grasping. A passive, non-actuated gripper further enhances the drone's ability to interact with objects in real-world scenarios. Experimental validations confirm the robustness and adaptability of the proposed approach, demonstrating its effectiveness in handling dynamic variations in mass and torque while maintaining stable flight.**

I. Introduction

Unmanned Aerial Vehicles (UAVs), particularly quadrotors, have gained significant attention in recent years, not only for their roles in surveillance and data collection [1], but also for more complex interactions with the environment. In particular, applications such as obstacle avoidance [2], object collection [3], and aerial manipulation have increased their interest in improving the UAVs' ability to physically interact with objects. These advances not only transform research fields, but also drive technological innovation in industries.

UAVs implemented in real world applications were originally designed to use integrated sensors and cameras for data capture, and were not intended for carrying external payloads [4]. As a result, when it comes to object manipulation tasks—such as picking up or transporting items—specialized drones and complex, often expensive grippers are typically required. This highlights the challenge of integrating payload handling capabilities into systems that were not initially designed for such purposes [5]. However, UAVs' versatility across various sectors, including security, manufacturing, and environmental monitoring, continues to expand [6].

Previous work has explored the development of both actuated and non-actuated grippers to enhance a drone's capacity for object handling. Actuated grippers, which provide control over force and pose, are widely used but introduce challenges related to system weight and complexity [7], [8]. Non-actuated designs, on the other hand, aim to reduce energy consumption and complexity, offering precise control with lower power requirements [9]. However, these systems still need to address the disturbances caused by external forces, such as the mass and torque variations encountered when grasping objects of different shapes and sizes.

One significant challenge when UAVs interact with objects is compensating for the disturbances that arise, especially when the object's mass or geometry causes unpredictable forces on the UAV. While earlier approaches have employed disturbance observers for compensating external forces during object manipulation [10], the impact of rotational dynamics on the UAV's stability during such tasks has been less explored. Notably, research has implemented systems such as the Uncertainty and Disturbance Estimator (UDE) [11], as well as various forms of Disturbance-Observer-Based Control (DOBC) [12], [13] to improve trajectory tracking and disturbance rejection in UAV systems. However, these methods primarily focus on either translational dynamics or rotational stability, with little emphasis on integrating both aspects into a unified framework.

The primary contribution of this paper is the development of a quaternion-based observer control system for multirotor UAVs performing aerial grasping tasks. This approach integrates two distinct observers (one for the translational and another for the rotational dynamics) thus allowing the system to act as fully actuated in both position and orientation. Building upon the approach presented in [14], this work uses an almost decoupling translational and rotational dynamics based on virtual fully actuated system through the use of a desired quaternion [15]. By utilizing quaternions for rotational dynamics, this method offers a singularity-free representation of the UAV's orientation, which aids in decoupling the rotational and translational dynamics, enhancing the overall stability of the UAV when interacting with objects. Furthermore, this new control structure improves the UAV's ability to reject disturbances from off-center grasps or objects with non-uniform shapes, allowing for more flexible and reliable aerial manipulation.

The remainder of the paper is organized as follows: In section II the quadrotor model and the quaternion-based control approach are presented. Section III is used to outline the dual-observer implementation. In section IV details on the experimental setup and results are introduced in graphs. Finally, in section V the implications of the results and suggested directions for future research are discussed.

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II. Quadrotor modeling and control

A. Quaternion dynamic model of the quadrotor

Based on the blade element theory, each rotor $i \in \{1,\dots,4\}$ in a symmetric quadrotor (see Fig. 1) generates a thrust $f_i := C_T \rho A_p r^2 \omega_i^2$ and a torque $\tau_i := C_Q \rho A_p r^3 \omega_i^2$ as a function of its angular velocity ω_i, the air density ρ, the blade area A_p, the rotor radius r, and the aerodynamic coefficients C_T and C_Q [16]. Thus, the total thrust force $\vec{F}_{th}$, acting vertically in the body reference frame, can be calculated as

$$\vec{F}_{th} := \begin{bmatrix} 0 \\ 0 \\ F_{th} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ f_1+f_2+f_3+f_4 \end{bmatrix},$$

and similarly, the total torque vector, acting in the same reference frame, can be expressed as

$$\vec{\tau} := \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} (-f_1-f_2+f_3+f_4)l\cos(\pi/4) \\ (-f_1+f_2+f_3-f_4)l\cos(\pi/4) \\ \sum_{i=1}^{4}\tau_i \end{bmatrix},$$

where l is the arm length of the quadrotor, extending from the center of mass to each motor.

Defining the state of the quadrotor as

$$X := \begin{bmatrix} \vec{p} \\ \dot{\vec{p}} \\ \boldsymbol{q} \\ \vec{\Omega} \end{bmatrix}, \tag{1}$$

where $\vec{p} \in \mathbb{R}^3$ is the position of the quadrotor in the inertial reference frame, $\dot{\vec{p}}$ is its velocity, $\boldsymbol{q} := q_0 + [q_1\ q_2\ q_3]^\intercal \in \mathbb{H}$ represents its orientation in quaternion form, and $\vec{\Omega} := [\omega_x\ \omega_y\ \omega_z]^\intercal$ is its angular velocity in the body reference frame. Thus, based on the expressions the total force and torque acting on the center of mass, the dynamic model of the quadrotor can be written as

$$\dot{X} = \frac{d}{dt}\begin{bmatrix} \vec{p} \\ \dot{\vec{p}} \\ \boldsymbol{q} \\ \vec{\Omega} \end{bmatrix} = \begin{bmatrix} \dot{\vec{p}} \\ \boldsymbol{q} \otimes \frac{\vec{F}_{th}}{m} \otimes \boldsymbol{q} + \vec{g} + \vec{\gamma}_F \\ \frac{1}{2}\boldsymbol{q} \otimes \vec{\Omega} \\ J^{-1}(\vec{\tau} + \vec{\gamma}_r - \vec{\Omega} \times J\vec{\Omega}) \end{bmatrix}, \tag{2}$$

where $\otimes$ represents the quaternion product, m and J are the mass and inertia matrix of the quadrotor, respectively, $\vec{g} := [0\ 0\ -9.81]^\intercal$ is the gravitational acceleration vector, and $\vec{\gamma}_F,\ \vec{\gamma}_r \in \mathbb{R}^3$ are unknown perturbation vectors.

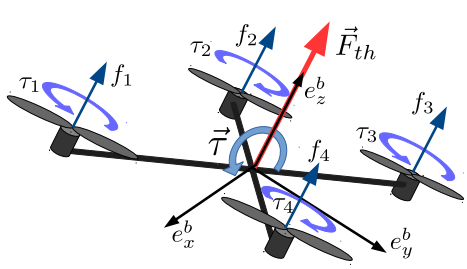

Fig. 1: Total thrust force $\vec{F}_{th}$ and torque $\vec{\tau}$ illustrated in [16].

B. Underactuation problem of the quadrotor

It is important to note that the dynamic system expressed in (2) is underactuated, with 6 degrees of freedom (3 for rotation and 3 for translation) and 4 control inputs (one for the total thrust force F_{th} and three for the torques $\vec{\tau} \in \mathbb{R}^3$). However, the rotational subsystem is fully actuated and can be written as

$$\dot{X}_r = \begin{bmatrix} \frac{1}{2}\boldsymbol{q} \otimes \vec{\Omega} \\ J^{-1}(\vec{\tau} + \vec{\gamma}_r - \vec{\Omega} \times J\vec{\Omega}) \end{bmatrix}, \tag{3}$$

According to [16] and [17], a quaternion logarithm mapping can be used to define an axis-angle rotational error between the quadrotor current orientation $\boldsymbol{q}$ and a desired rotational reference $\boldsymbol{q}_d$ as

$$\vec{\vartheta}_e := 2\ln(\boldsymbol{q}_d^* \otimes \boldsymbol{q}), \tag{4}$$

where $\boldsymbol{q}_d^*$ is the conjugate quaternion of $\boldsymbol{q}_d$, and $\vec{\vartheta}_e$ symbolizes the quadrotor orientation error expressed in its axis-angle notation. Considering small changes of $\boldsymbol{q}_d$, its derivative holds that

$$\dot{\vec{\vartheta}}_e = \frac{d}{dt}\vec{\vartheta}_e = \frac{d}{dt}2\ln(\boldsymbol{q}_d^* \otimes \boldsymbol{q}) \approx \vec{\Omega}. \tag{5}$$

Therefore, the rotational error subsystem can be rewritten as:

$$\dot{X}_{\vartheta_e} = \frac{d}{dt}\begin{bmatrix} \vec{\vartheta}_e \\ \vec{\Omega} \end{bmatrix} \approx \begin{bmatrix} \vec{\Omega} \\ J^{-1}(\vec{\tau} + \vec{\gamma}_r - \vec{\Omega} \times J\vec{\Omega}) \end{bmatrix}. \tag{6}$$

Defining the effects of the unknown rotational perturbations and the Coriolis term (which could vary in grasping tasks), as $\vec{\gamma}_\tau = \vec{\gamma}_r - \vec{\Omega} \times J\vec{\Omega}$, the rotational error equation is rewritten as:

$$\dot{X}_{\vartheta_e} \approx \begin{bmatrix} 0^{3\times3} & I^{3\times3} \\ 0^{3\times3} & 0^{3\times3} \end{bmatrix} X_{\vartheta_e} + \begin{bmatrix} 0^{3\times1} \\ J^{-1}(\vec{\tau} + \vec{\gamma}_\tau) \end{bmatrix}. \tag{7}$$

The first control's goal is to design a nominal controller stabilizing the system in its linear part and then, propose an observer that will compensate the nonlinearities components and the external perturbations. Thus, for the nominal controller the term $\vec{\gamma}_\tau \approx [0\ 0\ 0]^\intercal$ is neglected and therefore, the controller can be defined as:

$$\vec{\tau} = J\left(-2K_{pr}\ln(\boldsymbol{q}_d^* \otimes \boldsymbol{q}) - K_{dr}\vec{\Omega}\right), \tag{8}$$

where K_{pr} and K_{dr} are positive diagonal matrices containing the proportional and derivative gains, respectively.

Now, considering the translational subsystem

$$\dot{X}_t = \begin{bmatrix} \dot{\vec{p}} \\ \boldsymbol{q} \otimes \frac{\vec{F}_{th}}{m} \otimes \boldsymbol{q} + \vec{g} + \vec{\gamma}_F \end{bmatrix}, \tag{9}$$

the nominal controller (8) can ensure that $\boldsymbol{q} \to \boldsymbol{q}_d$ if $\vec{\gamma}_\tau$ is properly compensated. Rewriting from (2) the translational subsystem, it follows that

$$\dot{X}_t \approx \begin{bmatrix} \dot{\vec{p}} \\ \boldsymbol{q}_d \otimes \frac{\vec{F}_{th}}{m} \otimes \boldsymbol{q}_d + \vec{g} + \vec{\gamma}_F \end{bmatrix}. \tag{10}$$

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Using the methodology proposed in [16]-[15], a virtual control force $\vec{F}_u \in \mathbb{R}^3$ can be defined as

$$\vec{F}_u := \boldsymbol{q}_d \otimes \frac{\vec{F}_{th}}{m} \otimes \boldsymbol{q}_d + \vec{g},$$

so that the translational model becomes as follows:

$$\dot{X}_t \approx \begin{bmatrix} \dot{\vec{p}} \\ \vec{F}_u + \vec{\gamma}_F \end{bmatrix} = \begin{bmatrix} 0^{3\times3} & I^{3\times3} \\ 0^{3\times3} & 0^{3\times3} \end{bmatrix} X_t + \begin{bmatrix} 0^{3\times1} \\ \vec{F}_u + \vec{\gamma}_F \end{bmatrix}. \tag{11}$$

Since the control force $\vec{F}_u$ can be designed using any control strategy, the desired quaternion $\boldsymbol{q}_d$ is determined as the composition of two quaternions:

$$\boldsymbol{q}_d := \boldsymbol{q}_{xy} \otimes \boldsymbol{q}_z, \tag{12}$$

where $\boldsymbol{q}_{xy}$ represents the minimal rotation required to align the thrust vector $\vec{F}_{th}$ with the control force $\vec{F}_u$ [17], [15]. This quaternion is derived from the Euler-Rodrigues formula as:

$$\boldsymbol{q}_{xy} := \pm\sqrt{\frac{1+\frac{\vec{F}_u}{\|\vec{F}_u\|}\cdot\begin{bmatrix}0\\0\\1\end{bmatrix}}{2}} + \frac{\frac{\vec{F}_u}{\|\vec{F}_u\|}\times\begin{bmatrix}0\\0\\1\end{bmatrix}}{\left\|\frac{\vec{F}_u}{\|\vec{F}_u\|}\times\begin{bmatrix}0\\0\\1\end{bmatrix}\right\|}\sqrt{\frac{1-\frac{\vec{F}_u}{\|\vec{F}_u\|}\cdot\begin{bmatrix}0\\0\\1\end{bmatrix}}{2}}, \tag{13}$$

The second quaternion, $\boldsymbol{q}_z$, introduces a desired rotation around the z-axis (commonly referred to as the yaw angle), and it is computed as:

$$\boldsymbol{q}_z = \cos\left(\frac{\psi_d}{2}\right) + \begin{bmatrix}0\\0\\1\end{bmatrix}\sin\left(\frac{\psi_d}{2}\right), \tag{14}$$

where ψ_d denotes the target yaw orientation.

In order to establish a reference for evaluating control performance, a proportional-derivative (PD) state feedback controller has been used to regulate the translational dynamics of the quadrotor. Similarly as the rotational error model, a nominal force control $\vec{F}_u$ can be proposed under the assumption that external disturbances are negligible, i.e., $\vec{\gamma}_F \approx [0\ 0\ 0]^\intercal$. Then,

$$\vec{F}_u = -K_{pt}(\vec{p} - \vec{p}_d) - K_{dt}(\dot{\vec{p}} - \dot{\vec{p}}_d) - m\vec{g}, \tag{15}$$

where K_{pt} and K_{dt} are diagonal, positive-definite gain matrices responsible for the proportional and derivative actions, respectively. Here, $\vec{p}_d \in \mathbb{R}^3$ denotes the desired position vector, while m represents the mass of the quadrotor.

Notice from (15) that $\vec{F}_u = 0$ only when the quadrotor is on the floor, otherwise the mass is compensated and therefore (13) is also valid.

The process of computing the desired orientation and corresponding control inputs can be summarized in the following steps:

- **Compute the control force:** Determine the control force $\vec{F}_u$ using Eq. (15), which defines the translational dynamics based on the desired position and velocity.
- **Determine the intermediate quaternion:** Substitute the control force $\vec{F}_u$ into Eq. (13) to compute the intermediate quaternion $\boldsymbol{q}_{xy}$, representing the alignment of the thrust vector.
- **Incorporate the desired yaw angle:** Integrate the desired yaw angle ψ_d through Eq. (14) to obtain the complete target quaternion $\boldsymbol{q}_d$, which fully defines the desired orientation.
- **Apply the control torque:** Use Eq. (8) to apply the necessary control torque that ensures accurate tracking of the computed quaternion $\boldsymbol{q}_d$.
- **Calculate the total thrust:** Finally, compute the total thrust required to maintain the desired trajectory as:

$$F_{th} = \|\vec{F}_u\|. \tag{16}$$

III. Uncertainty and Disturbance Estimator

While the nominal controllers defined in Eqs. (8) and (15) assume an ideal environment without external disturbances and rely on precise, constant inertial parameters, real-world scenarios often present unpredictable perturbations. These disturbances can stem from environmental factors like wind gusts or, as explored in this work, variations in payload due to object grasping. To better capture these practical effects, the rotational and translational dynamics described by Eqs. (7) and (11) can be reformulated into two decoupled, linear systems with external disturbances as:

$$\dot{X}_{\vartheta_e} \approx A_{\vartheta_e} X_{\vartheta_e} + B_{\vartheta_e}\left[\vec{\tau} + \vec{\gamma}_\tau\right], \tag{17}$$

$$\dot{X}_t \approx A_t X_t + B_t\left[\vec{F}_u + \vec{\gamma}_F\right], \tag{18}$$

where $A_{\vartheta_e}{=}A_t{:=}\begin{bmatrix}0^{3\times3} & I^{3\times3}\\ 0^{3\times3} & 0^{3\times3}\end{bmatrix}$ and $B_{\vartheta_e}{=}B_t{:=}\begin{bmatrix}0^{3\times3}\\ I^{3\times3}\end{bmatrix}$.

A. Disturbance rejection controllers

The proposed control strategy involves estimating disturbance forces and torques using two observers based on the *Uncertainty and Disturbance Estimator* (UDE) approach [11]. Specifically, two estimations, $\hat{\vec{\gamma}}_\tau(t)$ and $\hat{\vec{\gamma}}_F(t)$, are designed to satisfy $\hat{\vec{\gamma}}_\tau(t) \to \vec{\gamma}_\tau(t)$ and $\hat{\vec{\gamma}}_F(t) \to \vec{\gamma}_F(t)$, respectively. These estimates are then incorporated into the controllers described in (8) and (15).

1) Rotational subsystem: The control input for the translational system is defined as:

$$\vec{\tau}(t) = J\left(-2K_{pr}\ln(\boldsymbol{q}_d^* \otimes \boldsymbol{q}) - K_{dr}\vec{\Omega} - \hat{\vec{\gamma}}_\tau(t)\right), \tag{19}$$

From the rotational subsystem (17), the reduced-order observer following the UDE design procedure yields

$$\vec{\gamma}_\tau(t) = B_{\vartheta_e}^+\left[\dot{X}_{\vartheta_e}(t) - A_{\vartheta_e}X_{\vartheta_e}(t) - B_{\vartheta_e}\vec{\tau}(t)\right], \tag{20}$$

where $B_{\vartheta_e}^+ = \begin{bmatrix}0^{3\times3} & I^{3\times3}\end{bmatrix}$ is the pseudoinverse of B_{ϑ_e}. Since (20) cannot be directly computed, the following estimation is proposed:

$$\begin{aligned}\dot{\hat{\vec{\gamma}}}_\tau(t) &= -\Gamma_{\vartheta_e}\hat{\vec{\gamma}}_\tau(t)\\ &\quad + \Gamma_{\vartheta_e}B_{\vartheta_e}^+\left[\dot{X}_{\vartheta_e}(t) - A_{\vartheta_e}X_{\vartheta_e}(t) - B_{\vartheta_e}\vec{\tau}(t)\right],\end{aligned} \tag{21}$$

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where $\Gamma_{\vartheta_e} \triangleq \text{diag}(\Gamma_{\vartheta_{e1}}, \Gamma_{\vartheta_{e2}}, \ldots, \Gamma_{\vartheta_{e6}})$ with $\Gamma_{\vartheta_{ej}} > 0$ for $j : \{1, \ldots, 6\}$. To avoid the term $\dot{X}_{\vartheta_e}(t)$, the following change of variable is applied

$$\hat{\xi}_{\vartheta_e}(t) = \hat{\vec{\gamma}}_\tau(t) - \Gamma_{\vartheta_e} B^+_{\vartheta_e} X_{\vartheta_e}(t). \tag{22}$$

Taking the derivative of (22) and introducing the system model results in the reduced-order observer:

$$\begin{cases} \dot{\hat{\xi}}_{\vartheta_e}(t) = -\Gamma_{\vartheta_e}\hat{\xi}_{\vartheta_e}(t) - \left(\Gamma^2_{\vartheta_e} B^+_{\vartheta_e} + \Gamma_{\vartheta_e} B^+_{\vartheta_e} A_{\vartheta_e}\right) X_{\vartheta_e}(t) \\ \qquad -\Gamma_{\vartheta_e}\vec{\tau}(t), \\ \hat{\vec{\gamma}}_\tau(t) = \hat{\xi}_{\vartheta_e}(t) + \Gamma_{\vartheta_e} B^+_{\vartheta_e} X_{\vartheta_e}(t), \\ \xi_{\vartheta_e}(0) = -\Gamma_{\vartheta_e} B^+_{\vartheta_e} X_{\vartheta_e}(0). \end{cases} \tag{23}$$

2) Translational disturbance rejection: An estimation $\hat{\vec{\gamma}}_F(t)$ is designed such that $\hat{\vec{\gamma}}_F(t) \to \vec{\gamma}_F(t)$ and then added to the controller as

$$\vec{F}_u(t) = -K_{pt}(\vec{p} - \vec{p}_d) - K_{dt}(\dot{\vec{p}} - \dot{\vec{p}}_d) - m(\vec{g} + \hat{\vec{\gamma}}_F(t)). \tag{24}$$

From (18), the reduced-order observer following the UDE design procedure yields

$$\vec{\gamma}_F(t) = B^+_t \left[\dot{X}_t(t) - A_t X_t(t) - B_t \vec{F}_u(t)\right], \tag{25}$$

where $B^+_t = \left[0^{3\times3} \;\; I^{3\times3}\right]$ is the pseudoinverse of B_t. Since (25) cannot be directly computed, the following estimation is proposed:

$$\begin{aligned} \dot{\hat{\vec{\gamma}}}_F(t) = &-\Gamma_t \hat{\vec{\gamma}}_F(t) \\ &+ \Gamma_t B^+_t \left[\dot{X}_t(t) - A_t X_t(t) - B_t \vec{F}_u(t)\right], \end{aligned} \tag{26}$$

where $\Gamma_t \triangleq \text{diag}(\Gamma_{t1}, \Gamma_{t2}, \ldots, \Gamma_{t6})$ with $\Gamma_{tj} > 0$ for $j : \{1, \ldots, 6\}$. To avoid the term $\dot{X}_t(t)$, we apply the change of variable

$$\hat{\xi}_t(t) = \hat{\vec{\gamma}}_F(t) - \Gamma_t B^+_t X_t(t). \tag{27}$$

Taking the derivative of (27) and introducing the system model results in the reduced-order observer:

$$\begin{cases} \dot{\hat{\xi}}_t(t) = -\Gamma_t \hat{\xi}_t(t) - \left(\Gamma^2_t B^+_t + \Gamma_t B^+_t A_t\right) X_t(t) \\ \qquad -\Gamma_t \vec{F}_u(t), \\ \hat{\vec{\gamma}}_F(t) = \hat{\xi}_t(t) + \Gamma_t B^+_t X_t(t), \\ \xi_t(0) = -\Gamma_t B^+_t X_t(0). \end{cases} \tag{28}$$

The control strategy is then completed by numerically computing $\hat{\xi}_{\vartheta_e}(t)$ and $\hat{\xi}_t(t)$, and introducing systems (23) and (28) into their respective control laws (19) and (24). Thus, all the information required by the observers depends on the system states and control inputs.

IV. Results

This section presents two stages of evaluation. First, numerical simulations are used to compare the performance of a classical PD controller under three conditions: a) `without` estimators, b) with a `translational UDE`, and c) with a `full UDE` (applied to both translational and rotational dynamics). In addition, the `Luenberger` estimator, as described in [18], is also included in the analysis. Second, experimental results are provided for two scenarios: one `without` estimator and the other using the `full` UDE. To simplify the analysis, a color code is used in this section to identify each simulation and experiment.

A. Numerical simulations

The simulations were developed in MATLAB R2024b with an Euler numerical integrator, with a step-size of 1 [ms], in a personal computer with an i7-13650HX processor and 32 GB of RAM. The physical parameters of the UAV were chosen to approximate real flying conditions under a controlled environment. The mass of the UAV is 0.405 [kg] and its inertia matrix is defined in (29).The initial conditions are defined as $\vec{p}(0) = [0\ 0\ 0]^\intercal$ [m], and $\mathbf{q}(0) = [1\ 0\ 0\ 0]^\intercal$.

$$J = \begin{bmatrix} 2098 & 63.577538 & -2.002648 \\ 63.577538 & 2102 & 0.286186 \\ -2.002648 & 0.286186 & 4068 \end{bmatrix} \mathrm{e}^{-6}\ [\mathrm{kg}\cdot\mathrm{m}^2] \tag{29}$$

The desired task corresponds to regulating the Cartesian position to $\vec{p}_d = \begin{bmatrix}1 & 2 & 2\end{bmatrix}^\intercal$ [m]. Also, a non-constant disturbance, in translational coordinates, is considered as $\vec{\gamma}_F = \begin{bmatrix}\sin(\omega_f\, t) & \cos(\omega_f\, t) & 2\sin(\omega_f\, t)\end{bmatrix}^\intercal$ with $\omega_f = 0.5$ [Hz] $\forall\, t \geq 10$ [s]. Using the feedback gains for the controller and observers shown in Table I, the following figures show a comparison among each simulation performing the same desired task.

TABLE I: Parameters for numerical simulations.

Simulation	K_{pr}	K_{dr}	K_{pt}	K_{dt}	Γ_{ϑ_e}	Γ_t
Without	40	8	2	1	-	-
Luenberger					2	0.5
Trans. UDE					-	0.5
Full UDE					15	1

As shown in Fig. 2a, the performance during the first seconds is the same for all. Once the disturbance is applied, at $t \geq 10$, the contribution of the observers is used in the control signals. In particular, the behavior of the full UDE helps to quickly compensate the error norm as presented in Fig. 2b, where the error norm from the other methods remains bounded but not fully compensated. The proposed rotational estimation proves effective the handling of non-constant disturbances along the time. This compensation is also shown in Fig. 2c, where the thrust produced by the full UDE reaches the required value faster to maintain the UAV closer to the desired position than the others.

A similar behavior can be seen in Fig. 3, where the desired positions are reached only under ideal conditions. However, when non-constant disturbances are applied, the performance decreases if observers are not used. While the translational UDE and Luenberger perform capability, the inclusion of rotational estimations (full UDE) improves the non-constant disturbances compensation, as shown in Figs. 3a-c. Even though the disturbances in the z-axis are around 50 % of the UAV's weight, the full UDE reduces the error more effectively than the translational UDE or Luenberger observer.

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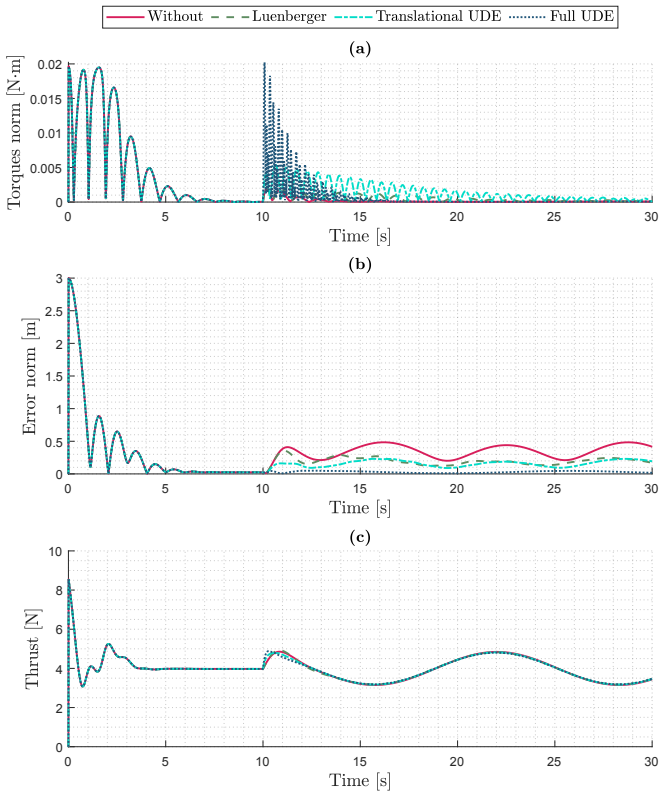

Fig. 2: General comparison of numerical simulations. **(a)** shows the torque norm, $||\vec{\tau}||$, whereas in **(b)** the translational error norm $||\vec{p} - \vec{p}_d||$ is presented. Finally, **(c)** depicts the applied thrust $\vec{F}_{th}$.

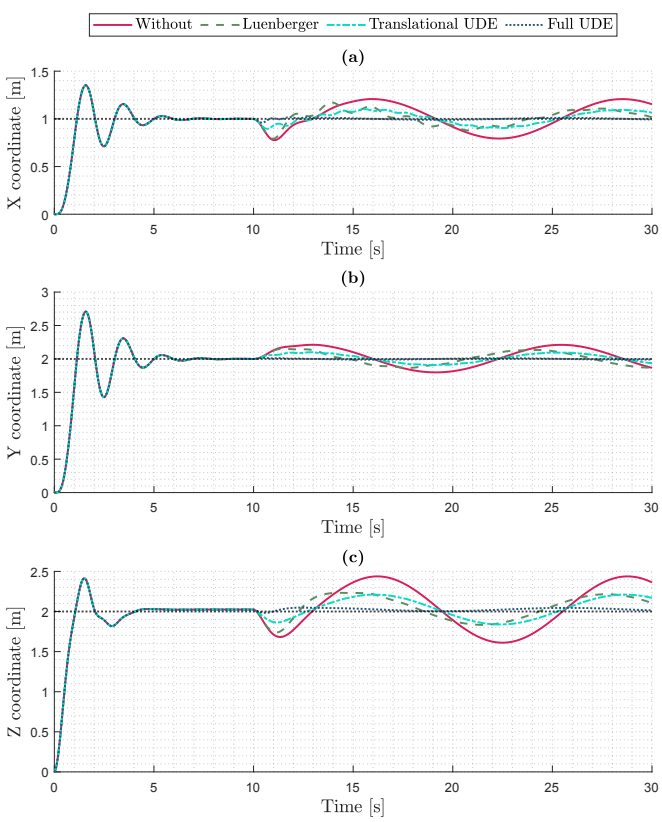

Fig. 3: Comparison of translational regulation to desired position (black dashed line). **(a)** illustrates the X coordinate, whereas the Y and Z coordinates are shown in **(b)** and **(c)**, respectively.

Fig. 4 highlights the advantages of using observers to handle disturbances. In particular, Fig. 4a shows that the accumulative error from the full UDE is the smallest, whereas the translational UDE and Luenberger behave in a similar way. Notice also that the full UDE achieves the error convergence along the time, even subject to non-constant disturbances, as shown in Fig. 4b. In contrast, the worst case corresponds to the case without observers, as it is expected, where the error convergence is not achieved. Likewise, the full UDE has the fastest response as depicted in Fig. 4c, whereas the others are continuously dealing with the disturbances even when they were applied a few seconds before. A summary of the metrics, as well as the error mean and its standard deviation, is presented in Table II.

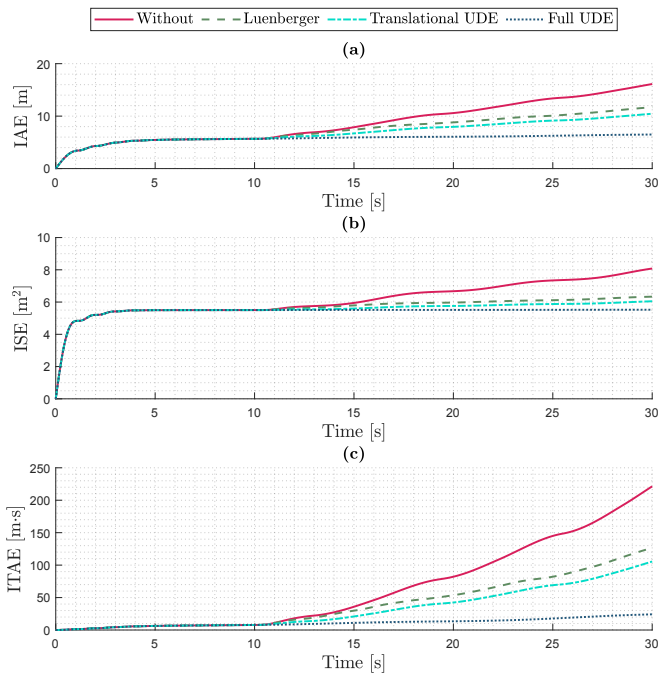

Fig. 4: Error metric analysis of numerical simulations. **(a)** depicts the Integral Absolute Error (IAE), **(b)** shows the Integral Squared Error (ISE), and **(c)** presents the Integral Time Absolute Error (ITAE).

TABLE II: Summary of simulation results.

Simulation	IAE [m]	ISE [m²]	ITAE [m·s]	Mean	Std. Dev.
Without	16.1198	8.0793	221.4028	**0.0066**	0.3419
Luenberger	11.6928	6.3339	126.3605	0.0165	0.3289
Trans. UDE	10.4555	6.0484	105.5268	0.0176	0.3241
Full UDE	**6.4931**	**5.5243**	**24.1687**	0.0266	**0.3184**

* The mean and standard deviation are given in meters.

B. Experimental results

Based on the numerical simulation results, the full UDE is expected to offer strong performance under non-constant disturbances. Therefore, two study cases were proposed to evaluate properly its behavior. The first study case corresponds to using only the PD controller `without` observers, as described in (8) and (15), whereas the

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second one includes the `full UDE`, based on (19) and (24). A video of the second experiment is available at: `https://youtu.be/z2qvN5186J4`.

These experimental results were obtained using Fl-AIR (Framework *libre* AIR) [19], an open-source C++ platform designed for deploying robotic applications, especially for UAVs. In Fig. 5, the general experimental setup is presented, where a Parrot AR.Drone 2.0 was equipped with a passive gripper mounted on its underside to grasp a target object. The *OptiTrack* [20] motion capture system is used to measure the position of the UAV and the target object, while the attitude in quaternions and angular velocities are given by the UAV's internal Inertial Measurement Unit (IMU). The onboard computer processes the OptiTrack and IMU data to compute the control signal, as well as the full UDE, to perform the desired task. Notice that the ground station only works as a monitor, which helps to reduce response time.

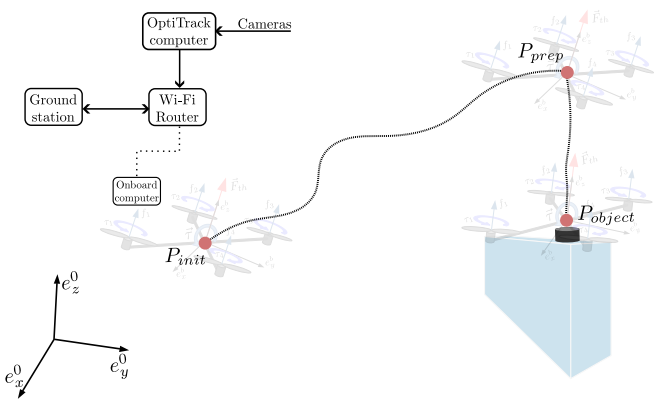

Fig. 5: Experimental setup and desired task. This trajectory was desgined to evaluate the UAV's behavior while it is transporting an object, which generates torques on each axis due to its own inertial matrix.

The proposed task consists of taking the UAV off to regulated to the initial position P_{init}. Once stabilized, the target position then is acquired to compute a quintic polynomial trajectory, which will be followed until reaching a preparatory position above the object P_{prep}, defined with a z-offset. Then, the UAV descends to take the object at P_{object} and subsequently returns to the preparatory position P_{prep}. Finally, the UAV transports the object to the initial position P_{init}, as depicted in Fig. 5. The use of a quintic polynomial trajectory ensures a smooth and feasible motion path that satisfies the constraints on position and velocity.

TABLE III: Parameters for experimental results.

Experiment	K_{pr}	K_{dr}	K_{pt}	K_{dt}	Γ_{ϑ_e}	Γ_t
Without	1.2	0.15	0.2	0.1	-	-
Full UDE					5	1

The controller and observer parameters used in the experiments are presented in Table III. These parameters were tuned based on the original weight of the UAV (without the gripper and the weight of the object), in order to properly evaluate the estimation capabilities and the real performance of the PD controller when the conditions change. The weight of the gripper is 0.04 [kg], and the object is 0.13 [kg], which is less than the recommended maximum payload [14].

Fig. 6a illustrates how the control torques present a bounded behavior in both experiments. However, the torques applied in the full UDE generally remain more consistent, as a consequence of the estimation. In contrast, the experiment without estimation requires greater variations in torque, which could imply a higher power demand. A similar situation occurs in Fig. 6b, where the error norm is continuously increasing given that the initial tuning is not adequate for the new mass and inertial matrix of the system if any observer is used. This is due to the controller being tuned for the original system parameters, which do not account for the added mass and the altered inertial matrix. In contrast, the full UDE correctly compensates these changes, maintaining performance even during the object manipulation phase, particularly during the ascent from P_{object} to P_{prep}, which could be a potentially critical point in the task. Notice that this estimation also impacts the thrust signal, where the full UDE maintains a nominal value even during the object grasping phase, as demonstrated in Fig. 6c.

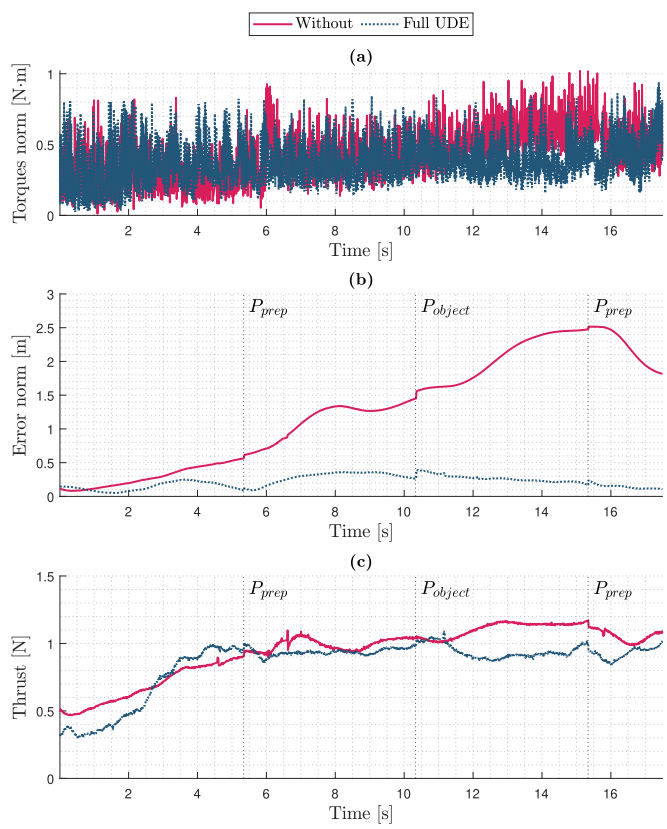

Fig. 6: General comparison of experimental results. The torque norm $||\vec{\tau}||$ is presented in **(a)**, the translational error norm $||\vec{p} - \vec{p}_d||$ is shown in **(b)**, and the applied thrust $\vec{F}_{th}$ is depicted in **(c)**.

Fig. 7a shows how the full UDE adequately handles the accumulative error, which highlights its robustness during all the grasping tasks performed. Similarly, it is capable of compensating for large errors that could directly affect the accuracy of the task, as Fig. 7b indicates, maintaining

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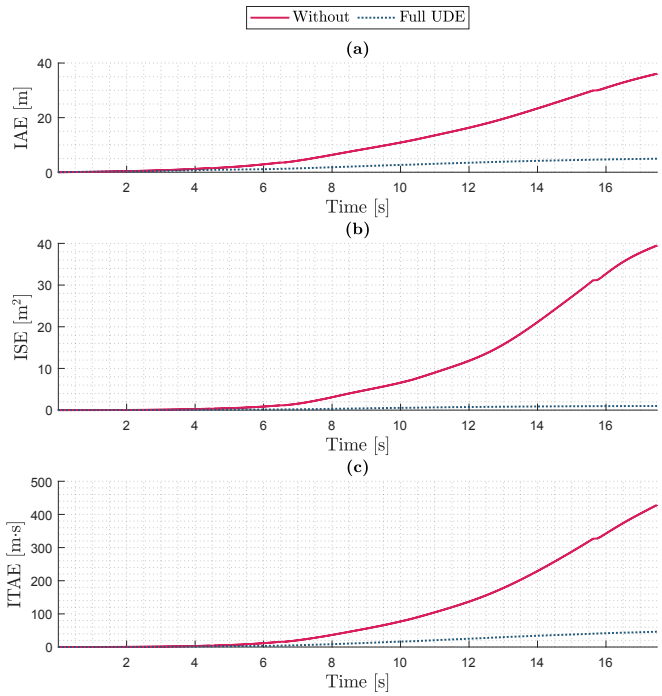

Fig. 7: Error metric analysis of experimental results. **(a)** depicts the Integral Absolute Error (IAE), **(b)** shows the Integral Squared Error (ISE), and **(c)** indicates the Integral Time Absolute Error (ITAE).

stable flight. Moreover, the quick response of the full UDE allows a reduction in the ITAE metric (Fig. 7c), ensuring good performance. Finally, a summary of the experimental results is presented in the Table IV.

TABLE IV: Summary of experimental results.

Experiment	IAE [m]	ISE [m²]	ITAE [m·s]	Mean	Std. Dev.
Without	36.0900	39.4965	428.5023	0.3382	1.75600
Full UDE	**4.9567**	**0.9799**	**46.2604**	**0.1080**	**0.2411**

* The mean and standard deviation are given in meters.

V. Conclusion and Future Work

This paper presented a full-state observer controller based on quaternions, which was validated through numerical simulations and experiments in grasping tasks. The numerical simulations demonstrated the advantages of using a quaternion-based observer to effectively compensate for unknown parameters and external, non-constant disturbances such as varying mass.

As a result, the proposed approach can be applied in various scenarios where traditional controllers may require re-tuning or prior knowledge about the task. Experimental results, obtained using a Parrot AR.Drone 2.0 in a grasping task, showed low cumulative errors, robustness, and faster responses, as indicated by the IAE, ISE, and ITAE metrics.

The findings of this paper encourage further research on improving controllers, particularly observers, to enhance the performance of traditional controllers under varying conditions. With advances in artificial intelligence, observers could integrate adaptive laws to use external information to improve task performance over time.

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